

EMBEDDED AND VLSI TRAINING IN DEMAND

The \$25-billion global embedded systems market is indicative of the immense opportunity that awaits the training providers in this field

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The widening gap between demand and supply of skilled professionals for embedded systems and very large-scale integration (VLSI) design reinforces the need for industry-specific training to aspirants in this field. Embedded systems are one of the key drivers of R&D offshoring to emerging countries like India and this market is currently pegged at around \$25 billion globally, growing 16 per cent annually. The huge embedded systems and VLSI market is indicative of the opportunity that awaits the training providers in this field.

"The training market is growing at an exponential rate as there are over 130 chip design firms present in India with a turnover of more than \$100 million. The vast pool of talent and growing domestic market have helped the country emerge as an important centre for embedded and VLSI design," says Vikas Kalra, director-marketing, CETPA.

According to Maaz Jukaku, director, Emtrexe, "The Indian market has just started expanding in terms of usage of embedded/electronic products at consumer levels and it is waiting to revolutionise. The number of embedded product engineers required to do the job is very high."

Embedded engineers trained in the latest technologies are always in demand."

—Thushar V. Nadh, Managing Director, rhydoLABZ

Considering that on an average 20,000 students sign up for embedded system courses in training institutes every year and around 50,000 technical students need help to continue their academic projects, Thushar V. Nadh, managing director, rhydoLABZ, estimates the size of the Indian embedded system and VLSI training market at Rs 1 billion and expects an annual growth rate of 10 to 15 per cent. This estimation is further supported by Vishwanath Raju, director-technical, Vector Embedded Systems. Taking into account the online training and e-learning options as well, Loganathan V., CEO, ISM, puts the embedded and VLSI training business in India at around Rs 2.4 billion.

An overview of the market

The training market can be defined as a component of the entire embedded industry knowledge value-chain. Dr K. Subbarangaiah, professor and director, Veda IIT, identifies the demand-generating components of this value chain as 'product design and design serv-

ices companies→SoC engineering and platform solution providers→Fabs and EMS providers→testing and packaging companies→consumer market.'

Training institutes and universities lie on the supply end. So Sahim Baker, director, Grey Technolab, expands the value chain as 'vendor training service provider/training industry→technology aspirant→employer.' Raju adds a middle component of government and industry associations that provide suitable environment for the training industry to grow.

The nature of training itself can be a parameter for classification. In such a case, it is worth considering Loganathan's simple approach: corporate trainings on specific skills, academic projects related orientation trainings, public-domain direct training, online training and e-learning-based training.

The topics of interest in embedded systems training are programming languages, microcontroller-based application development, application development for preconfigured embedded platforms, RTOS-based application development, embedded Linux development, digital signal processing, and device driver and FPGA-based embedded system development.

"The VLSI training market broadly consists of VLSI front-end design (which includes hardware design and modeling using HDLs) and design

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